

HOUSE PRICE PREDICTION USING MACHINE LEARNING

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Project Title: House Price Prediction Using Machine Learning

Tools Used: Python, Pandas, NumPy, Scikit-learn, Streamlit

Project Type: Machine Learning Regression Project

1. Introduction

House price prediction is an important task in the real estate industry. Machine learning techniques can help estimate house prices based on various features such as location, area, and number of rooms.

2. Problem Statement

The objective of this project is to predict the price of a house based on property features to assist buyers, sellers, and real estate professionals.

3. Objectives

- Perform data cleaning and preprocessing
- Conduct exploratory data analysis (EDA)
- Build a regression model for price prediction
- Evaluate model performance
- Deploy the model using Streamlit

4. Dataset Description

The dataset includes house features such as area, number of bedrooms, bathrooms, location, and price as the target variable.

5. Technologies Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Streamlit.

6. Data Preprocessing

Missing values were handled, categorical variables were encoded, and features were prepared to improve model performance.

7. Exploratory Data Analysis (EDA)

EDA included visualization of price distribution, correlation between features, and identification of key factors affecting house prices.

8. Model Building

Regression models such as Linear Regression, Decision Tree Regressor, and Random Forest Regressor were used to predict house prices.

9. Model Evaluation

The model was evaluated using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared score.

10. Streamlit Deployment

The trained model was deployed using Streamlit to allow users to input house features and receive price predictions.

11. Conclusion

This project demonstrates how machine learning regression techniques can be used to estimate house prices and support data-driven decisions.